

AFFILIATED TO TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY



A Major Project on

“Sawari Sewa”

A Major project report submitted in partial fulfillment of the requirements for the degree of
Bachelors of Science in Computer Science and Information Technology awarded by
Tribhuvan University.

Submitted by:

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Submitted to

Lumbini ICT Campus

Department of Computer Science and Information Technology

Gaindakot-2, Nawalparasi

January 2025

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ACKNOWLEDGMENT

We would like to express our sincere gratitude to all those who have contributed to the development of this project proposal for a Sawari Sewa. This project is prepared in partial fulfillment of the requirement for the bachelor's degree in Computer Science and Information Technology as a part of our syllabus. We have made an effort in this project. However, it would not have been possible without the help and support of supervisor and coordinator.

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We would like to appreciate our friends who have directly and indirectly helped us in developing this project. Our motivation is to create a solution that meets their needs efficiently and effectively. We are committed to building a platform that enhances user experience and accessibility in the car rental market.

Thank you all for your support and contribution to this project. Any kind of suggestion or criticism will be highly appreciated.

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ABSTRACT

The rapid growth in demand for flexible and efficient vehicle rental services has necessitated the development of a robust digital solution. This project aims to design and implement a web-based vehicle rental management system, **Sawari Sewa**, to streamline operations and enhance customer experience. The proposed system facilitates seamless vehicle rentals by providing real-time vehicle availability, online booking, and secure payment options. It empowers administrators to efficiently manage vehicles, staff, and rental transactions, thereby increasing operational efficiency and customer satisfaction. By offering hassle-free and affordable rental options for various durations daily, weekly, or longer **Sawari Sewa** ensures a transparent, reliable, and enjoyable rental experience. The system is designed with user-centric methodologies, focusing on simplicity, reliability, and scalability. This document details the project's objectives, methodologies, system architecture, implementation strategies, and testing protocols, ensuring a comprehensive approach to delivering a state-of-the-art solution for the vehicle rental industry

Keywords: *Vehicle Rental Services, System Architecture, Affordable Rental Options, Hassle-free Rentals, Customer Satisfaction*

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CHAPTER 1: INTRODUCTION

1.1 Background

Welcome to Sawari Sewa. We are thrilled to offer a wide range of rental services to our valued customers, designed to meet your specific requirements and preferences. Our services include vehicle rental and further other can be updated. Vehicle rental services are increasingly becoming an essential part of urban transportation. They provide flexibility, convenience, and cost-effective solutions for various needs, including travel, business, and leisure. The traditional process of renting a car is often involving manual booking and limited accessibility. By using modern web technologies, this project aims to create an efficient and user-friendly platform that caters to the needs of customers and providers.

At our Service, we understand the importance of convenience and affordability, which is why we offer flexible rental plans that allow you to rent our vehicles for a day, a week, or fortnight.

1.2 Problem statement

Renting a Vehicles can often be hard for customers due to several factors such as limited availability of cars, high rental rates, hidden fees, complex reservation processes, unclear policies and terms, unresponsive customer support, and vehicle maintenance issues. Traditional car rental systems rely heavily on manual operations, which can lead to inefficiencies such as delays, limited customer reach, and errors in booking management. Customers often face challenges in locating available vehicles, making online payments, and receiving prompt service. Similarly, service providers struggle to manage their fleet effectively and maintain customer satisfaction. This project seeks to address these challenges by developing a web-based platform that automates and streamlines the car rental process.

By addressing these challenges, the car rental service can improve customer satisfaction, increase customer loyalty and retention, and enhance its brand reputation, leading to sustainable growth and profitability.

1.3 Objectives

Our objective is to provide the service of vehicles online for the different users for a day program or the tours. The primary goal of this project is to design and develop a comprehensive car rental system. Specific objectives include:

- Deliver reliable, well-maintained rental vehicles tailored to customer needs, with flexible plans for daily, weekly, or extended rentals.
- Ensure outstanding customer service through friendly, knowledgeable staff committed to helping customers find the ideal vehicle.
- Provide a seamless rental experience with online booking, easy payment options, and convenient pickup/drop-off locations, free from hidden fees.
- Continuously enhance services based on customer feedback to offer a transparent, hassle-free, and enjoyable rental experience.

1.4 Scope and Limitation

1.4.1 Scope

Our company has a significant opportunity to expand its scope in the market by continuously improving its services based on customer feedback and suggestions. Our company good render our service regularly according to the customer feedback and suggestion our company has great chance to grow its field in the market. The framework will enable real-time booking for a vehicle, tracking the availability of the vehicle, secure online transactions, and user-friendly navigation for customers and administrators. It will also allow customers to set up accounts and edit profiles.

1.4.2 Limitation

The initial version of the system will be focused on a single geographic location, with plans for expansion in future iterations. The system will require consistent internet connectivity, which may limit its usability in areas with poor network infrastructure.

- **Limited availability:** Depending on the location and time of year, rental vehicle may be in high demand, which can limit availability. This can be especially true during peak travel seasons or in popular tourist destinations.
- **Vehicle restrictions:** Some car rental companies may have restrictions.

1.5 Development Methodology

The Agile development methodology will be used for this project, promoting iterative development and regular feedback integration. This approach ensures continuous improvement and alignment with user needs.

An Agile approach could help ensure that all requirements are taken into account and addressed in the development process.

MVC (Model view and Controller) is the software development pattern which is widely used.

- **Model:** The Model manages the core data and business logic of the system, such as vehicles, users, and bookings. It interacts with the database to perform CRUD operations and enforces rules for valid rentals.
- **View:** The View handles the user interface (UI), presenting the data managed by the Model to the user. It ensures a seamless and user-friendly experience for renters and admins.
- **Controller:** The Controller acts as a bridge between the Model and the View, handling user requests, updating the Model, and returning responses to the user via the View.

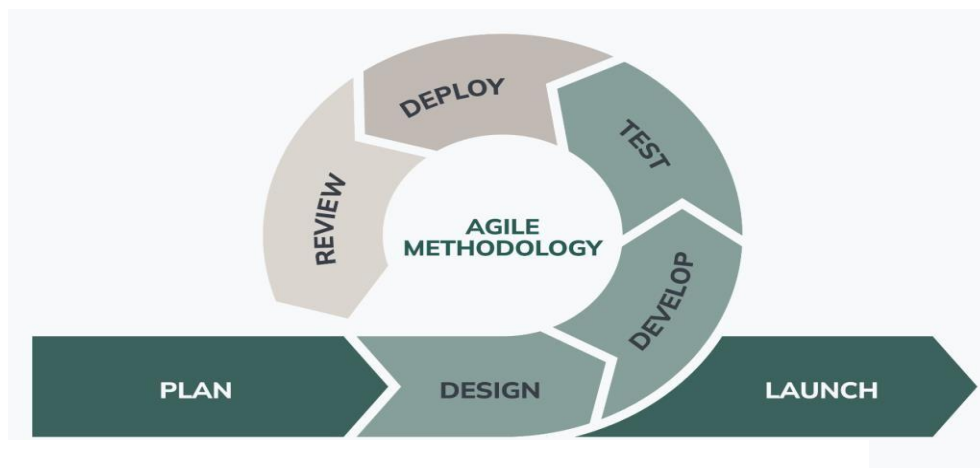


Figure 1 Agile Methodology

1.6 Report Organization

The report of the project is based on following format:

Chapter 1: Introduction: This is the primary section of the project where overview, problem statement, objectives, scope and limitations and, methodology has been discussed.

Chapter 2: Background Study and Literature Review: In this section, the descriptions of fundamental theories from intellectual personnel and general concepts related to the project has been discussed.

Chapter 3: System Analysis: This section describes the general architecture and flow mechanism of the system. It also describes the hardware and software needed to build the system and also feasibility analysis for the system.

Chapter 4: System Design: This section describes the diagrammatic description of the system. At first, whether the system is structured or object-oriented is identified then suitable diagrams to describe the system are designed here. Activity diagram, use case diagram, state diagram describes the entities involved in the system and the relation between them.

Chapter 5: Implementation and Testing: This part defines the tools used and the implementation details of the project. It also consists of different test cases used to test different units of the project.

Chapter 6: Conclusion and Future Recommendations: This section describes the progress made till to date and the future plans of the project.

CHAPTER 2: BACKGROUND & LITERATURE REVIEW

2.1 Background study

Online Vehicle rental services like Sawari Sewa offer flexible, convenient, and affordable transportation solutions to a diverse customer base, including tourists, business travelers, and those needing temporary vehicles. The industry has evolved significantly, transitioning from traditional brick- and-mortar models to digital platforms and innovative services. While facing competition from ride-sharing services and changing consumer preferences, car rental companies are adapting to these challenges by embracing technology and offering tailored solutions to meet the evolving needs of their customers. Some of previously built systems similar to our proposal idea are: -

- <https://rentinnepal.com/>
- <https://sparkcar.org/>
- <https://www.rentalnepal.com/>
- <https://cartipur.com/>
- <https://letsdrive.com.np>

2.2 Literature Review

It is an important transportation option for many travelers, offering a flexible and convenient way to rent a vehicle for a specific period of time. In recent years, the industry has faced several challenges, including increased competition from ride-sharing services and changing consumer preferences. In response, car rental companies have adopted new technologies and business models to stay competitive and meet the evolving needs of their customers. The literature review will focus on scholarly articles, industry reports, and case studies covering:

- Consumer Behavior: Understanding customer preferences and expectations.
- Service Quality Metrics: Evaluating factors influencing customer satisfaction.
- Competitor Analysis: Comparison of traditional vs. disruptive entrants (e.g., peer-to-peer vehicles sharing).
- Future Prospects: Review of emerging trends like autonomous vehicle rentals and blockchain integration.

CHAPTER:3 SYSTEM ANALYSIS

3.1 System Analysis

System analysis is a critical process in developing and troubleshooting systems, ensuring systems their effective implementation and alignment with user needs. For the development of Sawari Sewa, this process encompasses observing end-user interactions, evaluating system components, and analyzing methodologies to enhance system functionality. Key activities include assessing source code to understand the agile methodology applied in software development, conducting feasibility studies to validate the project's viability, and reviewing comparable systems to identify best practices. Additionally, sentiment analysis may be integrated to gauge user satisfaction and feedback for continuous improvement.

3.1.1 Requirement Analysis

Requirement analysis is a vital phase in the system analysis process, focused on defining the expectations and needs of the users for the proposed application. This involves gathering, documenting, and validating requirements to ensure the system delivers value to its users and aligns with its objectives.

3.1.1.1 Functional Requirements

Functional requirements define the specific capabilities and behaviors the system must exhibit to meet user expectations. For the Sawari Sewa Service, the functional requirements are as follows:

- **User Authentication:** Users can log in to their accounts using a secure combination of email and password.
- **Vehicle Search and Display:** The system will provide a dynamic list of available vehicles based on the user's search criteria, such as vehicle type, vehicles model and so on.
- **Online Booking:** Users can select a preferred vehicle and rental period, and confirm their booking through an intuitive online interface.

- **User Information Management:** Users can update or delete their personal information within the system, ensuring data accuracy and privacy.

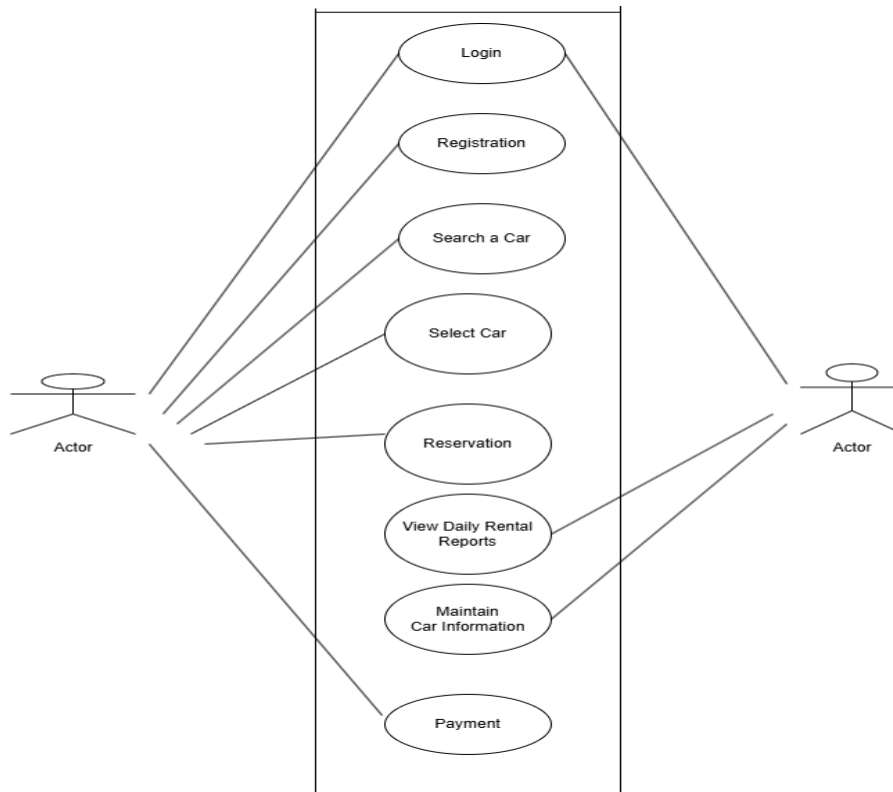


Figure 2 Use Case Diagram

The above use case diagram defines the context of the system. The users are able to search for car and view details of the car. The users are able to get recommendations based on their input. They are also able to view reviews of other users and get sentiments. The system processes the data, calculates similarities and generates recommended car to the user. It also performs sentiment analysis and shows reviews and sentiments to the user.

3.1.1.2 Non-Functional Requirements

Non-functional requirements are the characteristics or attribute of the system that can judge its operation. The non-functional requirement of the system includes:

- **Performance:** The system should be able to handle a high volume of users and transactions without slowing down or crashing. The system response time should be fast and reliable.

- **Security:** The system should be secure and protect user data from unauthorized access or breaches. The system should use encryption to protect user passwords and payment information.
- **Usability:** The system should be user-friendly and easy to navigate. The system interface should be intuitive and accessible to all users, including those with disabilities.
- **Scalability:** This system is scalable to provide many features.

3.1.1 Use Case Diagram of the system

Feasibility analysis is a crucial step in the system development process that evaluates whether the proposed system is viable and worth pursuing. It examines the practicality of implementing the system from various perspectives to ensure its success. For Sawari Sewa service, the feasibility analysis focuses on technical, operational, economic, and

3.1.2.1 Technical Feasibility

Technical feasibility assesses whether the technology, tools, and expertise required to build the system are available and sufficient. The proposed system relies on frameworks for web and mobile application development, secure authentication protocols, and reliable database management systems. With advancements in cloud computing and APIs for payment processing and GPS integration, the system can be built and scaled effectively. Assess the availability of necessary technologies such as mobile app development frameworks (React Native, Flutter), web development frameworks (Angular, React), cloud platforms (AWS, Azure, GCP), and payment gateways.

3.1.2.1 Operational Feasibility

Operational feasibility examines the system's ability to meet the organization's goals and satisfy user needs. Sawari Sewa is designed to provide a seamless experience, from searching for vehicles to making bookings. By ensuring user-friendly interfaces and efficient workflows, the system can meet customer expectations and support operational efficiency for the business. Implement a 24/7 customer support system, including phone, email, and chat support. Develop a rigorous vehicle maintenance schedule to ensure optimal performance and customer satisfaction

3.1.2.2 Economic Feasibility

Economic feasibility evaluates the cost-effectiveness of the project by comparing estimated development and operational costs against potential benefits. Costs include software development, hosting, and maintenance, while benefits include increased revenue through online bookings and improved customer retention. A return-on-investment (ROI) analysis indicates that the system's benefits outweigh its costs.

Table 1 Sample Economic Feasibility

Month	Revenue	Expenses	Profit/ Loss
1	Rs 40,000	Rs 35,000	Rs 80000
2	Rs 40,000	Rs 45,000	Rs 90000

3.1.1.1 Legal Feasibility

Legal feasibility focuses on compliance with relevant laws and regulations, such as data protection, consumer rights, and transportation licensing. The system will adhere to data privacy standards (e.g., GDPR or local regulations), ensure secure handling of payment information, and comply with local laws governing vehicle rentals and insurance. Such as business licensing, Employment Laws, Contract Law

3.1.1.2 Schedule Feasibility

Schedule feasibility evaluates whether the project can be completed within the allocated time frame. By adopting an agile development methodology, the system can be built iteratively, allowing for flexibility and timely delivery of key features. This ensures that the project timeline aligns with organizational goals and market demands.

By addressing these aspects, the feasibility analysis confirms that the rental service is a viable and sustainable project, capable of meeting user needs and achieving business objectives while adhering to technical, operational, economic, and legal requirements. It is the total time schedule for the development and release of final system

Table 2 Schedule for Project

Project Sawari Sewa (Grantt Chart)	Start Date	End Date	Progress
proposal	01/05/24	01/16/24	80%
Research and design	01/16/24	02/29/024	70%
prototype 1	02/29/2024	03/29/2024	8%
Test and Further plan	04/05/2024	04/25/24	80%
prototype 2	04/26/24	05/02/24	500%
Test and Further plan	08/05/24	08/01/24	300%
System Maintaining	08/02/24	08/15/2024	30 %
Final documentation		12/24/24	40%

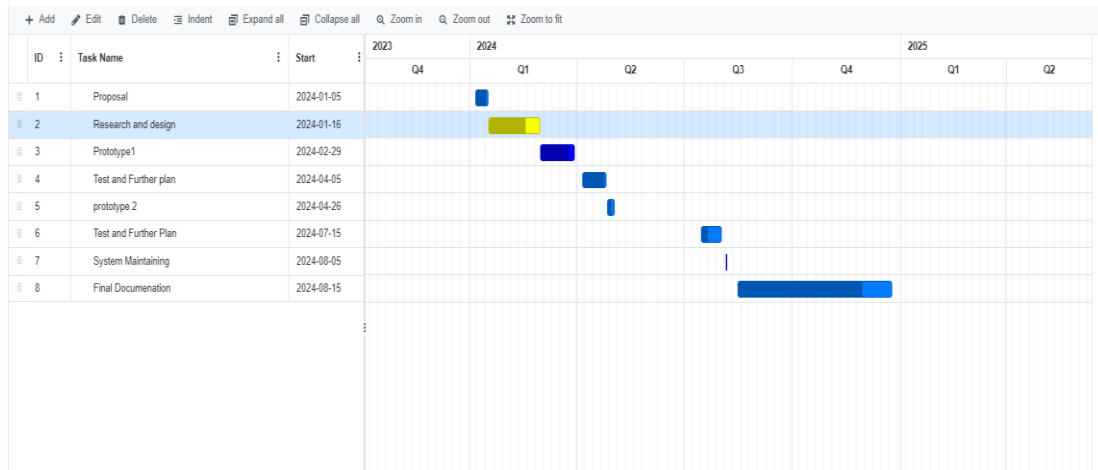


Figure 3 Grantt Chart

3.1.2 Analysis

3.1.3.1 Process Modeling

Process modeling is a critical step in system analysis, allowing for a clear, graphical representation of how data flows through the system. It outlines how the system captures, manipulates, stores, and distributes data between its components and external entities. One effective tool for process modeling is the Data Flow Diagram (DFD), which visualizes the processes, data stores, and external interactions in a structured manner. DFDs help in identifying system boundaries, key processes, and data flow dynamics, providing a foundation for system design and implementation.

CHAPTER 4: SYSTEM DESIGN

4.1. Design

Design is a critical phase in software development, serving as the blueprint for how the system will meet its goals and function effectively. It requires meticulous planning and decision-making to ensure that the software achieves its objectives while maintaining efficiency and reliability. A well-thought-out design minimizes errors, optimizes system performance, and reduces the likelihood of costly revisions during later stages of development. Poor design can result in increased processing time, slower response times, and excessive workload, ultimately impacting user satisfaction and system effectiveness.

4.1.1 Class Diagram

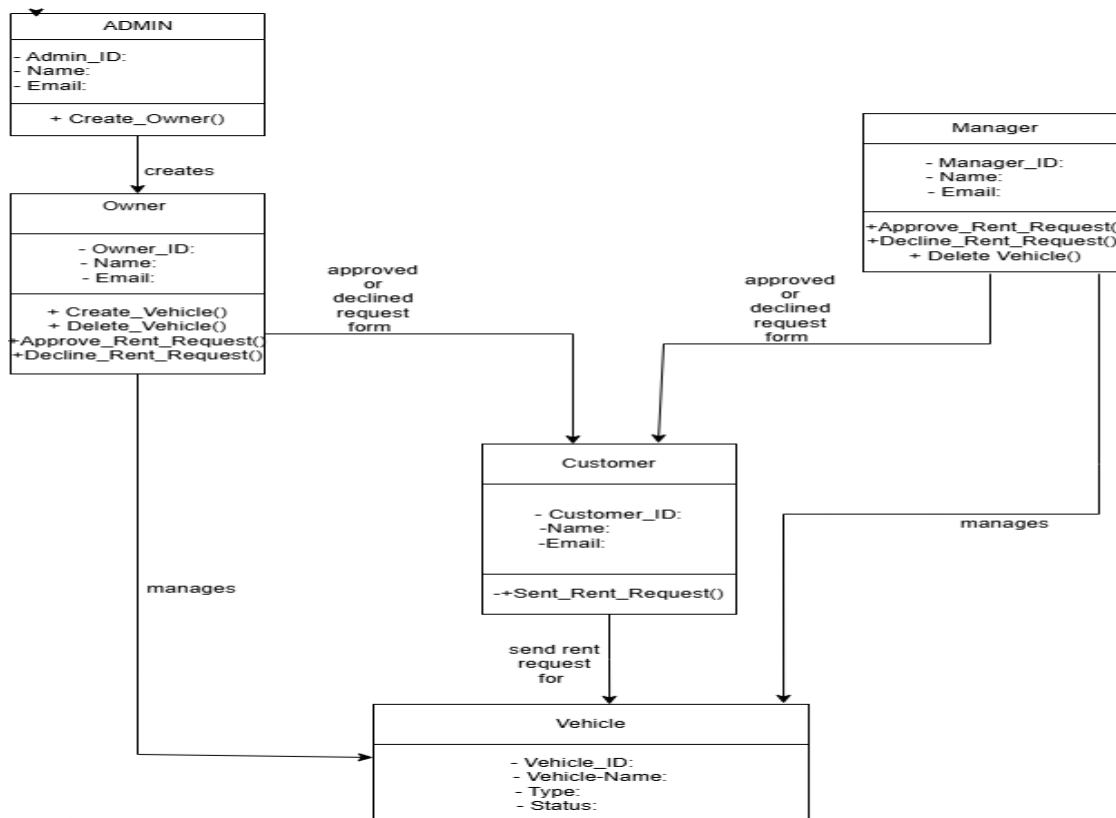


Figure 5 Class Diagram of Online Vehicles Renting System

4.1.2 Sequence Diagram

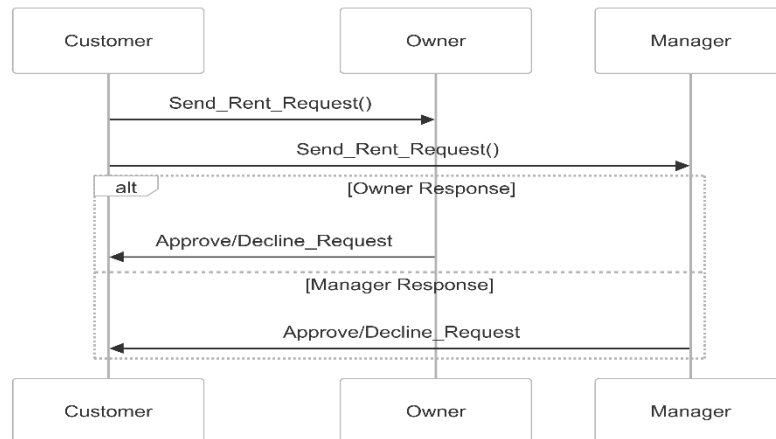


Figure 6 Sequence Diagram

This figure represents a **sequence diagram** for a vehicle rental request process in the "Sawari Sewa" system. It illustrates the interaction between three key entities: **Customer**, **Owner**, and **Manager**. The process begins with the **Customer** sending a rental request (`Send_Rent_Request()`) to the **Owner**, who forwards it to the **Manager** for approval. Both the **Owner** and the **Manager** respond to the request with either an approval or a decline (`Approve/Decline Request`), completing the communication cycle back to the **Customer**. The diagram highlights the decision-making hierarchy and ensures that rental requests are processed systematically.

4.1.3 Schema Diagram

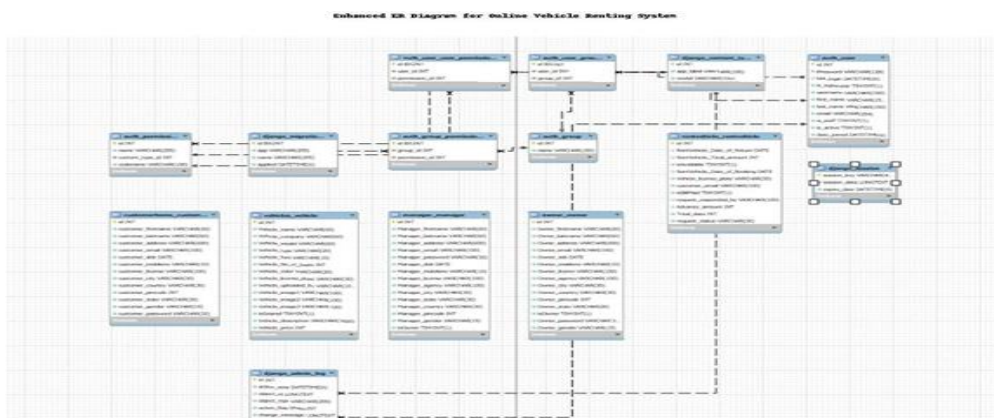


Figure 7 Schema diagram of Online Vehicle Renting System

This enhanced Schema diagram for the Online Vehicle Renting System represents the key entities and their relationships. It includes tables for users, vehicles, bookings, payments, and feedback, highlighting how data is interconnected. The design ensures seamless management of vehicle inventory, customer bookings, payment processing, and user interactions. Relationships between entities such as customers and bookings, vehicles and availability, and payment records reflect a well-structured database system for efficient operations.

4.1.3 Interface Design

Following are some screenshots of the system in different phases.

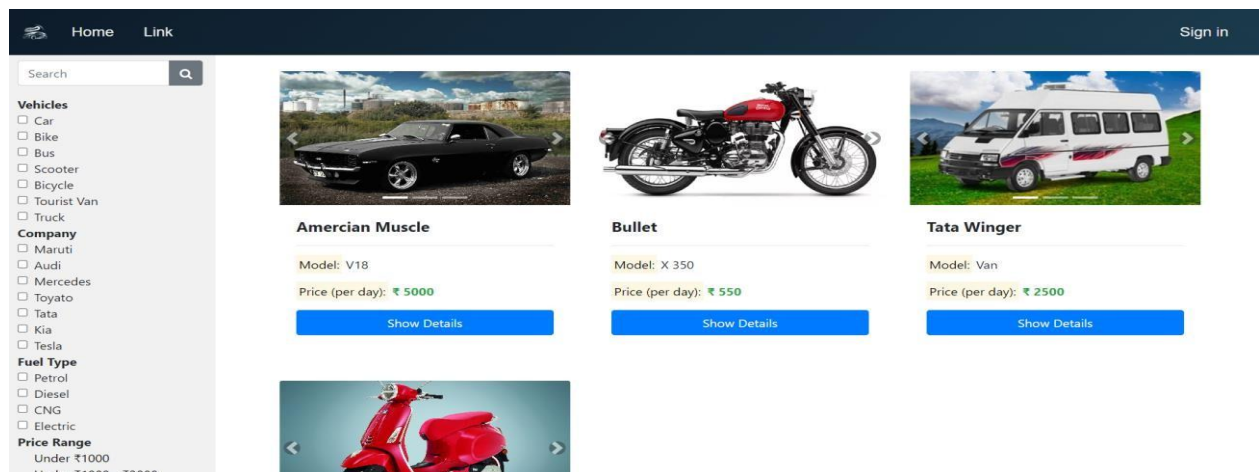


Figure 8 UI of Homepage

Sent Requests

All

Pending

Accepted

Declined

#	Vehicle	Booking Date	Return Date	Price (per day)	Number of Days	Total Amount	Request Status
1	Land Cruiser VXR V8!	Jan. 17, 2025	Jan. 20, 2025	3500	4	14000	Pending Cancel

Figure 9 Request Checking by Customer

4.1.4 Incremental Model

For this project, we adopted the **Incremental Model**. This approach divides the development process into manageable cycles or iterations, making it ideal for situations where requirements are initially unclear. Each iteration was refined based on stakeholder feedback, enabling a flexible and responsive development process.

The incremental model proved particularly effective for this project due to its adaptability. It allowed for early launches and rapid updates, which are advantageous for small projects with brief life cycles.

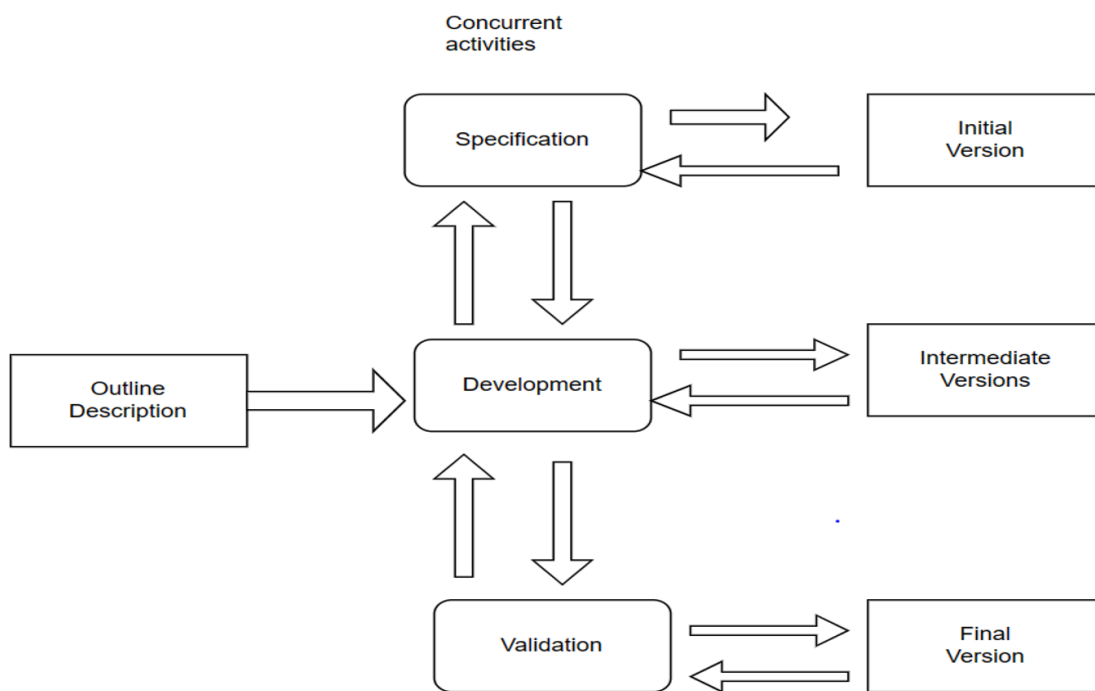


Figure 10 Incremental model

4.2 Algorithm Details

Below are the descriptions of the algorithms that are include in the system:

1. Interval Overlap Check for Vehicle Availability

In a vehicle rental system, one of the critical tasks is to ensure that a vehicle is available for a customer on the requested dates. The Interval Overlap Check algorithm is used to determine if a vehicle has already been rented out during the customer's requested date range. This check ensures that no two rental bookings overlap for the same vehicle.

Algorithm Overview: The system stores the booking date ranges for each vehicle in the database. When a customer requests to rent a vehicle for a specific date range, the system compares the requested date range with the existing bookings for that vehicle. If the requested date range overlaps with any of the existing bookings, the request is declined. If no overlap is found, the booking is accepted. This algorithm ensures that the rental process is smooth and avoids double- booking of vehicles.

2. Greedy Approach for Rental Request Processing

The Greedy Approach is applied to process rental requests and allocate available vehicles to customers. This algorithm follows a simple approach: it aims to accept the first available vehicle for the customer's requested dates and ensures that the vehicle is assigned efficiently.

Algorithm Overview: The system checks if the vehicle is available for the requested date range using the Interval Overlap Check. If the vehicle is available, the system approves the booking request and updates the vehicle's status as rented.

3. Pagination and Efficient Data Retrieval

When dealing with a large number of vehicles or rental records, retrieving and displaying all the data at once can be inefficient and slow. The Pagination and Efficient Data Retrieval algorithm helps in splitting large datasets into smaller, manageable chunks, improving the performance and responsiveness of the system.

Algorithm Overview: Pagination limits the number of records displayed per page, making the user experience smoother and faster.

CHAPTER 5: IMPLEMENTATION & TESTING

5.1 Implementation

Implementation is a process of ensuring that the information system is operational. It involves constructing a new system from scratch or constructing a new system from the existing one. The implementation section provides a brief description of technologies used for this implementing project.

5.1.1 Technologies Used

1. Tailwind CSS: Tailwind CSS is the CSS framework which helps in designing faster and helps to make a design more beautiful. It helps to provide a quick implementation of the styles.

2. JavaScript: JavaScript is one of the most simple, versatile and effective languages used to extend functionality and interactivity in websites. JavaScript is executed on the client side. JavaScript is relatively fast to the end user as the code is executed on the user's computer.

3. Python: Python is a versatile and widely-used programming language known for its simplicity, readability, and vast ecosystem of libraries. It is an ideal choice for developing software systems due to its ability to handle a wide range of tasks efficiently, including web development, data processing, and machine learning.

4. Django: Django is a high-level Python web framework that facilitates the rapid development secure and maintainable websites. It is widely used for its scalability, built-in features, and adherence to the DRY (Don't Repeat Yourself) principle.

5.2 Testing

5.2.1 Unit testing

Unit Testing was done to test field validations, navigation, functionality of the programs and their blocks. These tests are applied on various functions within each program and other critical program blocks. Table below outlines two sample test cases for Unit Testing performed on the system:

5.2.1.1 Unit Testing on Registration

Table 3 Unit Testing on Registration

Test Case	Expected Outcome	Result
Users enter required information (such as name, email, password, phone number, and other details)	1. Register the users. 2. Verify that user is home page. 3. Login with the registered email and password.	Successful
Users forget to enter one information	An error message must be display	Successful

5.2.1.2 Unit Testing

Table 4 Unit Testing

Test case	Expected outcome	Result
Users enter required information (such as name, email, password, phone number, driving license photo and other details)	1. Verify that driver is redirected to the home page. 2. Email Link is sent manually to verify driver. 3. Login with the registered email and password.	Successful
Users forget to enter one information	An error message must be display	Successful

5.2.2 Integration Testing

Table 5 Integration Testing

	Test Case	Expected Outcome	Result
Vehicle availability	Choose type of required, vehicle and select rent start date and end date	Display the result the vehicle may available on that date	Successful
Booking	Users click on Sent request	Then the request is seen by manager or owner. Decision is based on owner and manager	Successful
Multiple date	User click on the Sent request	If the vehicle is already rented on that date, then the notification is display, Vehicle is not Available for these dates!!	Successful

5.2.3 System Testing

System testing for a vehicles rental service involves testing the different components and functionalities of the system to ensure that it meets the requirements and functions as expected. Some of the essential system testing scenarios for a vehicles rental service include user registration, vehicles reservation, availability checking, rental check-in/out, payment processing, and reporting and analytics. These tests aim to verify that users can successfully create an account, reserve a vehicle, check-in/out rental vehicle, make payments, and recognize

CHAPTER 6: CONCLUSION & REFERENCE

6.1 Conclusion

The vehicle rental business has transitioned from traditional, location-dependent operations to a more dynamic, internet-driven model. This evolution has made renting vehicles more accessible and convenient for customers. With features like online reservations, doorstep delivery, and in-office pickup, the system ensures a seamless and user-friendly experience. By leveraging technology, the vehicle renting system addresses modern customer needs, enhancing efficiency and satisfaction while setting a new standard for the industry.

6.2 Future Scope

We aim to elevate our vehicle renting system by integrating advanced features designed to enhance user experience and operational efficiency. Our plans include incorporating secure and seamless payment gateways, such as Stripe, PayPal, and Razor pay, to facilitate hassle-free transactions. To foster real-time user engagement, we will introduce features like live chat support and instant booking updates, ensuring an interactive and responsive platform. Furthermore, we plan to expand accessibility by developing an Android application, enabling users to access our services conveniently on mobile devices. Leveraging advanced methodologies, such as the latent factor model, we will analyze user preferences to deliver personalized recommendations and services. These enhancements, along with additional user-friendly features, will position our platform as a cutting-edge solution in the vehicle rental industry.

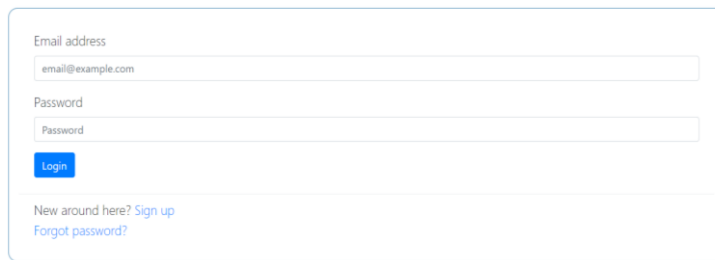
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Appendices

- **Login**

Login



A login form with a light blue border. It contains two input fields: 'Email address' with the placeholder 'email@example.com' and 'Password' with the placeholder 'Password'. Below the password field is a blue 'Login' button. At the bottom, there are two links: 'New around here? Sign up' and 'Forgot password?'.

Figure 11 Login Page

- **After Login**

Login



A login form similar to Figure 11, but with the input fields filled. The 'Email address' field contains 'sub_cst2077@lct.edu.ng' and the 'Password' field contains '*****'. The blue 'Login' button is still present. The links 'New around here? Sign up' and 'Forgot password?' are at the bottom.

Figure 12 After login

- **Vehicles List**

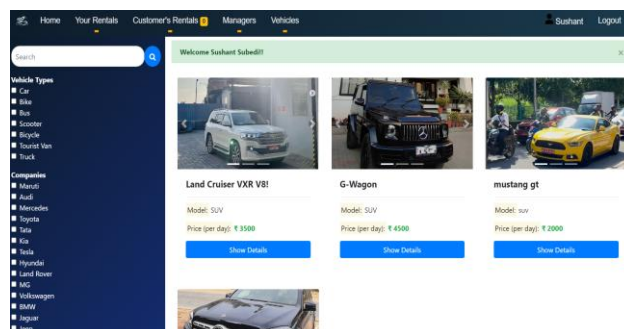


Figure 13 UI home page

- **Vehicles Detail**



Check Availability

Name	Mercedes GLS 350D
Company	Mercedes
Model	SUV
License Plate	1000
Fuel Type	Diesel
Seats	5
Gear Type	Auto
About	The overall feel is simple, and restrained, yet massively confident, with a subtle colour palette, large squared-off elements defined by brushed, it is a very complete car. It stocks a stocking great engine with strong power, it's as

Figure 14 Vehicles Details

- **Vehicles Slot Checking**



January 2025

↑

↓

Mo	Tu	We	Th	Fr	Sa	Su
30	31	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
3	4	5	6	7	8	9

Clear

Today

Model	SUV
License Plate	1000
Fuel Type	Diesel
Seats	5
Gear Type	Auto
About	The overall feel is simple, and restrained, yet massively confident, with a subtle colour palette, large squared-off elements defined by brushed. It is a very complete car. It stocks a stocking great engine with strong power, it's as extremely comfortable .
Price	₹3200 (per day)
Delete Vehicle	

Figure 16 Vehicles Slot Checking

